HTML & CSS INTERNSHIP PROJECT

Personal Portfolio Website

Project Documentation

Built with HTML5 and CSS3 | Hosted on GitHub Pages

SUBMITTED BY

Suhana Thabassum

B.Tech CSE (AI/ML)

HTML & CSS Internship Project

LIVE WEBSITE

https://suhana-thabassum.github.io/

Table of Contents

1. Abstract 3

2. Introduction 3

3. Project Objectives 3

4. Software and Technologies Used 3

5. System Requirements 4

6. Website Design and Structure 4

7. Implementation 5

8. Project Screenshots 7

9. Testing and Results 10

10. Advantages and Learning Outcomes 11

11. Limitations and Future Enhancements 11

12. Conclusion 12

13. References 12

1. Abstract

This document describes the **Personal Portfolio Website** created by Suhana Thabassum, a B.Tech CSE (AI/ML) student, as part of an HTML & CSS internship project. The website introduces the author, summarises her academic background and technical skills, presents twelve programming projects as cards with links to their GitHub repositories, and provides a Contact section with a LinkedIn QR code.

The site is built from two files, `index.html` and `style.css`, using HTML5 and CSS3 only, and is published through GitHub Pages at suhana-thabassum.github.io. It uses a dark theme with a responsive layout that adapts from desktop to mobile screens. All descriptions in this report are based on the submitted source files and screenshots.

2. Introduction

A **personal portfolio website** is a web page, or a small set of pages, where a person presents who they are, what they know and what they have built. Unlike a resume, which is a short static document, a portfolio lets visitors see real work, follow links to source code and contact the owner directly.

For students seeking internships, a portfolio is useful because it:

- shows practical ability through projects instead of only listing skills;
- gives recruiters one shareable link that gathers education, skills, projects and contact options;
- demonstrates basic web-development skills such as page structure, styling and responsive design;
- is easy to update as the student learns new technologies.

3. Project Objectives

- Showcase personal information, education, technical skills, projects and contact details.
- Build a responsive website using HTML5 and CSS3 that works on desktop and mobile screens.
- Provide access to the author's professional profile through a LinkedIn QR code.

Coverage in the final website: personal information (Home, About), education (one sentence in About), skills (four cards), projects (twelve cards) and contact options (LinkedIn button, GitHub link, QR code). No email address or phone number is displayed.

4. Software and Technologies Used

Technology	Purpose in this project
HTML5	Page structure and content: semantic elements, links, headings and the QR image.
CSS3	All visual design: colours, typography, Flexbox and Grid layouts, hover effects, animations and media queries.
Visual Studio Code	Code editor used to write and organise the project files (as stated by the author).
Google Chrome	Browser used to view the site; the screenshots show Chrome Developer Tools device emulation.
GitHub and GitHub Pages	Version control and free static hosting; the live site is served from the github.io address.

5. System Requirements

The website is a static site with no server-side code, so the requirements are modest.

	For developing the website	For viewing the website
Hardware	Computer with 4 GB RAM (8 GB recommended), 1 GB free storage, 1366 x 768 display or better	Any laptop, tablet or smartphone with a modern screen
Operating system	Windows 10/11, macOS or Linux	Windows, macOS, Linux, Android or iOS
Software	Visual Studio Code, a modern web browser (e.g. Google Chrome), Git (for GitHub)	A current web browser such as Chrome, Edge, Firefox or Safari
Network	Internet connection for GitHub and for pushing changes	Internet connection to open the live site

6. Website Design and Structure

The portfolio is a single page. A sticky navigation bar at the top links to five sections, and a footer closes the page. The table below summarises each part as it appears in `index.html`.

Section	Content in the website
Navigation bar	Logo "Suhana." and five links: Home, About, Skills, Projects, Contact. It stays at the top while scrolling.
Home	Welcome label, heading "Hello, I'm Suhana Thabassum", subtitle "B.Tech CSE Student | AI/ML Enthusiast", short introduction, two buttons (Explore My Projects, Contact Me), three tags (Python, AI/ML, Problem Solving) and a code-window illustration with two floating badges.
About	Three short paragraphs: the author is a B.Tech CSE (AI/ML) student at Malla Reddy College of Engineering and Technology, Hyderabad; she builds projects in Python, Java, HTML and CSS; and she is seeking internship experience.
Skills	Four cards: *Programming* (Python, Java, C Basics); *Web Development* (HTML, CSS, Flask); *Data & Databases* (NumPy, Pandas, SQL, CSV); *Core Concepts* (DSA, OOP, AI/ML, Git & GitHub).
Projects	Twelve numbered project cards (01-12), each with a title, short description, technology tags and a "View on GitHub" link.
Contact	Message about internship opportunities, a "Connect on LinkedIn" button, a link to the GitHub profile and the LinkedIn QR code with the caption "Scan to connect on LinkedIn".
Footer	Logo, the line "Designed and built with HTML & CSS." and a 2026 copyright notice.

6.1 Project cards

The twelve project cards are listed below exactly as they appear on the website.

No.	Project	Tags shown on the card
01	Student Performance Analytics System	Python, NumPy, Pandas
02	TRAQ (traffic congestion forecasting)	Python, FastAPI, ML
03	QUIZLY (quiz web application)	Python, Flask, Web App
04	MEDEASE (prescription assistant)	TypeScript, AI
05	MRCET College Website	HTML, CSS
06	My Portfolio	HTML, CSS
07	Area of Shapes - Java	Java, OOP
08	Contact Book Manager	Python, JSON
09	Expense Tracker	Python, CSV
10	Python To-Do List	Python, File Handling
11	LinkedIn QR Code Generator	Python, QR Code
12	OTP Generator	Python, Random

6.2 Colour theme and typography

The site uses a dark theme: a very dark navy page background (`#090d18`) with near-white text (`#f2f5ff`) and softer blue-grey text for descriptions. Accent colours are purple, blue and cyan, applied as gradients on the main buttons (`#7761f7` to `#428ff3`) and on highlighted words in the headings (`#9d8aff` to `#55d9ef`). Cards use dark, slightly transparent gradient backgrounds with thin light borders and rounded corners. Text is set in the *Segoe UI* font with Arial and a generic sans-serif as fallbacks.

6.3 Navigation and responsive layout

Each navigation link points to a section `id` (`#home, #about, #skills, #projects, #contact`), so clicking a link scrolls to that section. On wide screens the Home section is split into two columns (text and illustration) and the Skills and Projects sections use two-column grids. On screens narrower than 650 px every grid becomes a single column and the navigation bar stacks the logo above the links. A tablet breakpoint at 850 px adjusts spacing and sizes.

7. Implementation

7.1 Role of the two files

`index.html` (about 516 lines) holds the content and structure of the page. `style.css` (about 860 lines) controls how that content looks and how it adapts to different screen sizes. The HTML links to the stylesheet with one line in the `<head>`:

```
<link rel="stylesheet" href="style.css">
```

7.2 HTML structure

The page starts with `<!DOCTYPE html>` and `<html lang="en">`. The `<head>` contains the character set, a viewport tag (needed for correct display on phones), a meta description and the page title "Suhana Thabassum | Developer Portfolio". The `<body>` is organised with semantic elements:

```
<header class="navbar">  ... logo + <nav> with 5 links ...  </header>
<main>
    <section id="home">      hero text, buttons, code-window illustration
    <section id="about">     about card
    <section id="skills">    four skill cards
    <section id="projects">  twelve <article> project cards
    <section id="contact">   LinkedIn button, GitHub link, QR image
</main>
<footer class="footer"> ... </footer>
```

- **Semantic tags** (`header, nav, main, section, article, footer`) describe the purpose of each area, which helps browsers and screen readers.
- **Accessibility hints** include `lang="en"`, an `aria-label` on the navigation, and descriptive `alt` text on the QR image.
- **External links** (14 in total, including the GitHub project links) open in a new tab with `rel="noopener noreferrer"`.
- **No JavaScript file** is used. The only script-like code is a small inline `onerror` attribute on the QR image that shows a text message if the image fails to load.

7.3 CSS techniques used

The table lists techniques that are present in `style.css`, with a plain-language explanation of each.

Technique	Where it is used	What it does
Reset and `box-sizing`	`* { margin:0; padding:0; box-sizing:border-box }`	Removes browser default spacing so sizes are predictable.
Flexbox	Navigation bar, buttons, tags, skill badges, project cards, footer	Places items in a row or column and wraps them when space runs out.
CSS Grid	Hero, About card, Skills grid, Projects grid, Contact card	Creates two-column layouts, e.g. `repeat(2, minmax(0, 1fr))`.
Sticky positioning	`.navbar`	Keeps the navigation bar visible while the page scrolls.
Gradients and `background-clip`	Buttons, heading highlights, cards	Smooth colour blends, including gradient-coloured text.
Transitions and hover effects	Nav links, buttons, skill cards, project cards, project links	Cards and buttons lift slightly and change colour on hover.
Keyframe animations	Code window, floating badges, blinking cursor	Gentle floating and blinking movement.
Media queries	`max-width: 850px`, `max-width: 650px`, `prefers-reduced-motion`	Adapt the layout to tablets and phones; reduce motion for users who prefer it.
`clamp()`	Headings and hero title	Lets font sizes scale smoothly with screen width.
Smooth scrolling	`html { scroll-behavior: smooth; scroll-padding-top }`	Animates jumps to sections and keeps headings clear of the sticky bar.
Pseudo-elements	`body::before`, `body::after`	Create soft coloured glows in the page background.

Example 1 - Grid layout and the mobile breakpoint (from `style.css`):

```
.projects-grid {
    display: grid;
    grid-template-columns: repeat(2, minmax(0, 1fr));
    gap: 24px;
}

@media (max-width: 650px) {
    .projects-grid { grid-template-columns: 1fr; }
}
```

On wide screens the project cards sit in two columns. When the screen is 650 px wide or less, the rule inside the media query replaces it with a single column.

Example 2 - Hover effect on project cards:

```
.project-card:hover {
    transform: translateY(-7px);
    border-color: rgba(132, 151, 255, 0.56);
    box-shadow: 0 18px 45px rgba(68, 79, 176, 0.12);
}
```

When the pointer moves over a card, it lifts by 7 px and its border and shadow change. The `transition` property on `.project-card` makes this movement smooth.

8. Project Screenshots

Figures 1 and 2 are the screenshots supplied by the author. Figures 3 to 5 were produced for this document by rendering the submitted `index.html` and `style.css` in a headless Chrome browser, so that sections not visible in the supplied screenshots can also be shown.

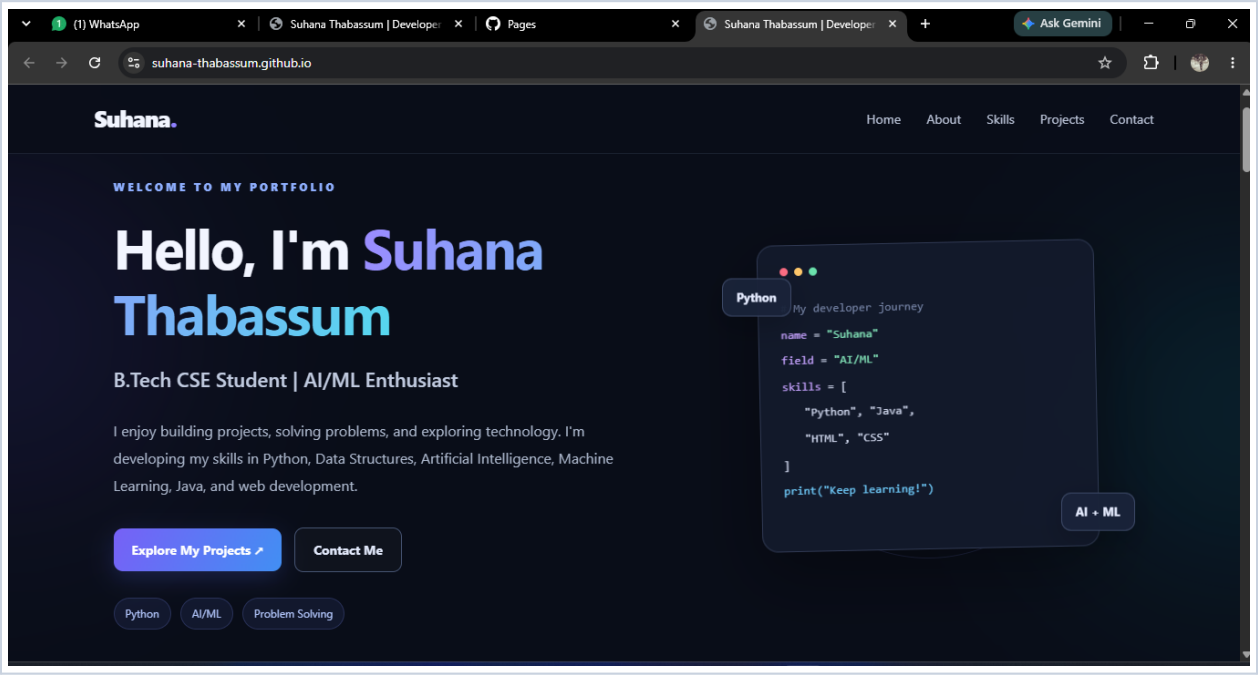


Figure 1: Desktop View - Home section on the live site (suhana-thabassum.github.io)

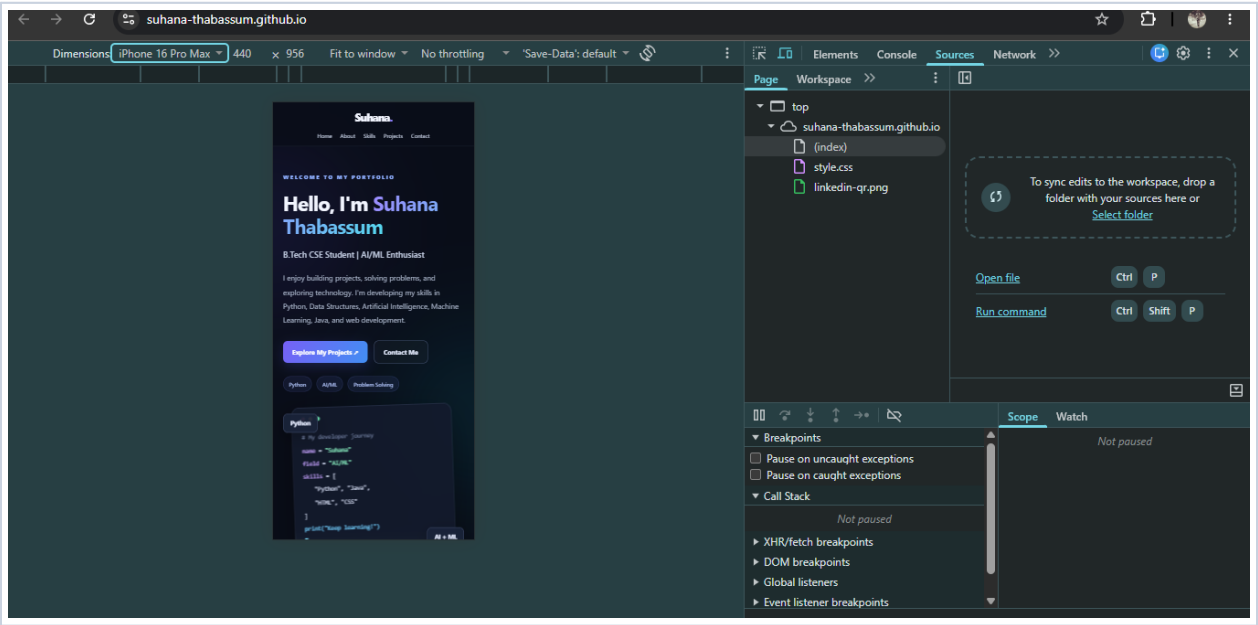


Figure 2: Mobile View - Chrome Developer Tools device emulation (iPhone 16 Pro Max, 440 x 956). The Sources panel lists index, style.css and linkedin-qr.png served from the live site.

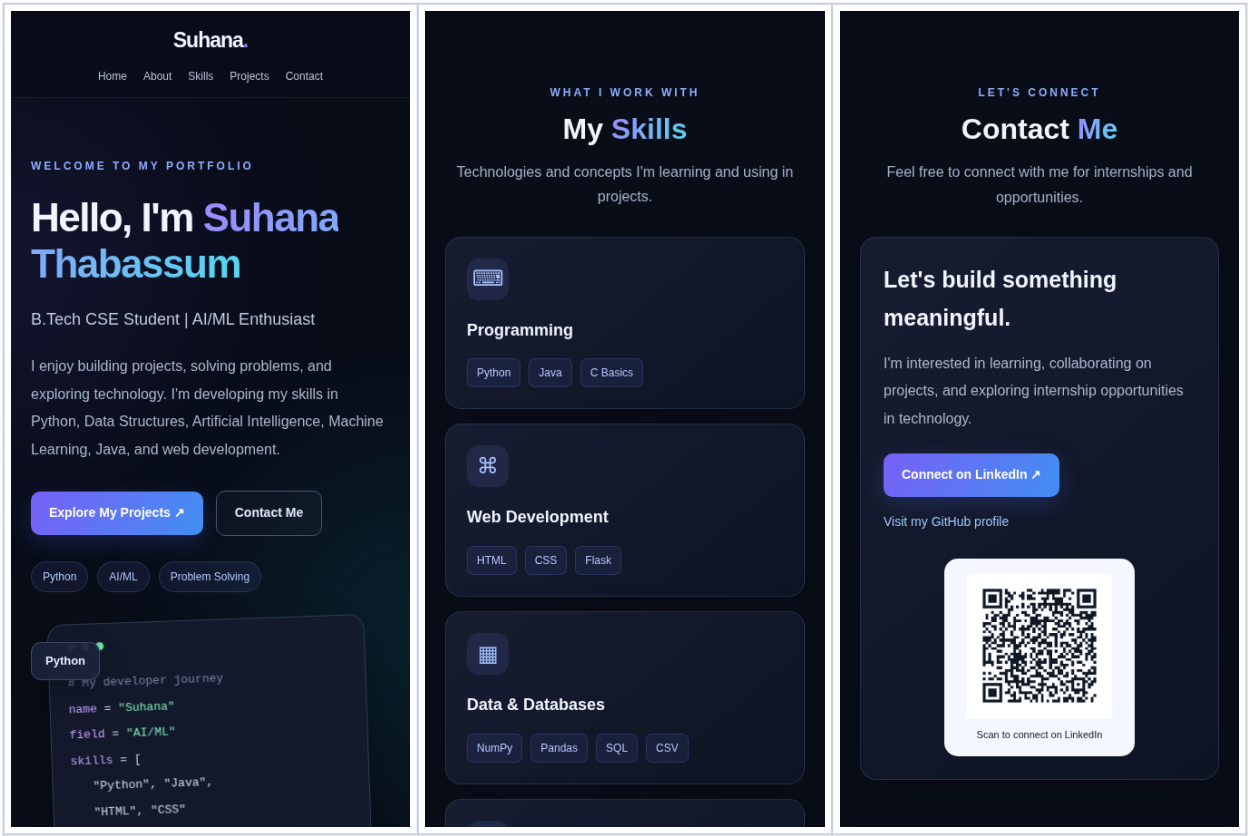


Figure 3: Mobile layout rendered at 440 px width - Home, Skills and Contact sections in a single column

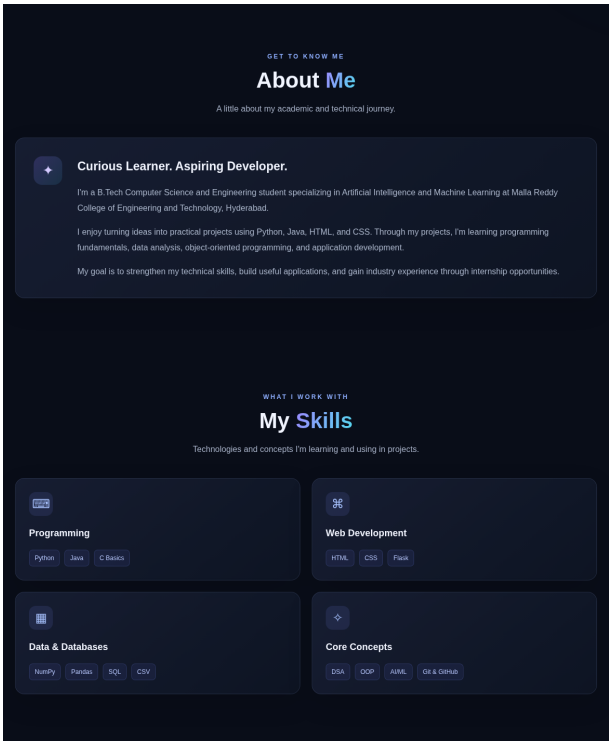


Figure 4: Desktop - About and Skills sections (1366 px render)

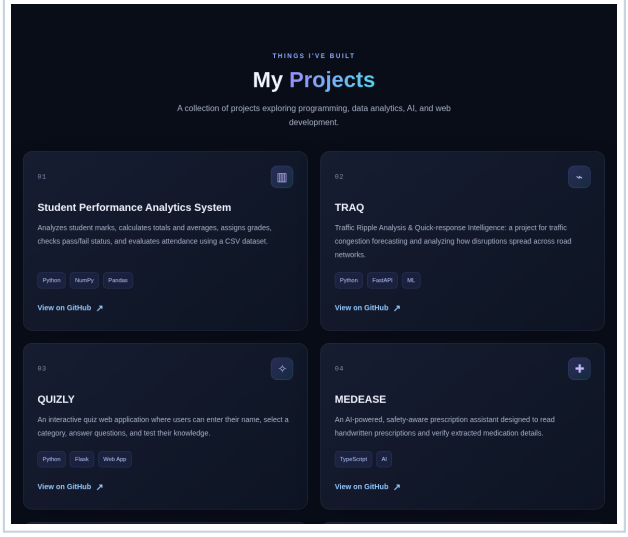


Figure 5: Desktop - Projects section, first four of twelve cards (1366 px render)

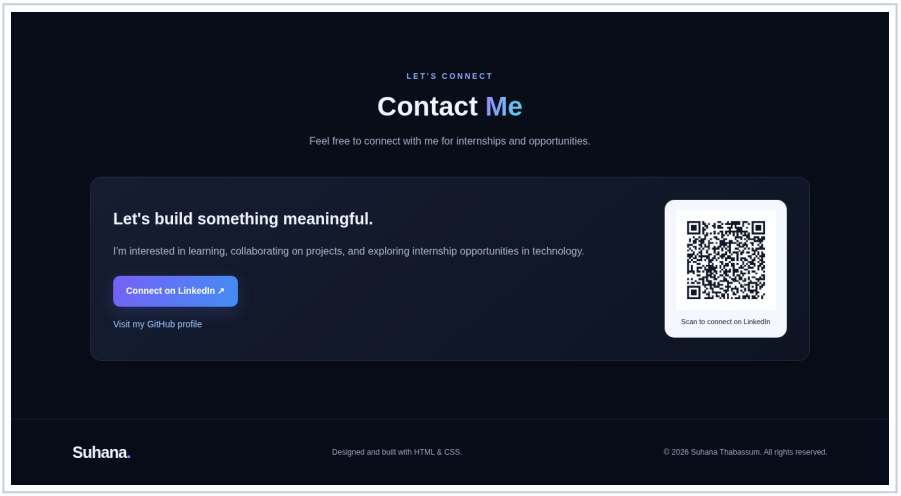


Figure 6: Contact section with the LinkedIn QR code (`linkedin-qr.png`) and footer

LinkedIn QR code. The QR image is displayed in the Contact section (Figure 6) so that a visitor can scan it with a phone camera. When the supplied `linkedin-qr.png` was decoded during preparation of this document, it resolved to a LinkedIn profile address, so it is appropriate to include it in the report as part of the Contact section.

9. Testing and Results

The table records what could be confirmed from the submitted files, the screenshots and checks run while preparing this document (a headless Chrome render at four screen widths, a script that checks links and section IDs, and a QR-code decode). Items that could not be confirmed are marked **Not verified** rather than assumed to pass.

Test case	Expected result	Evidence	Status
Site opens on GitHub Pages	Page loads at suhana-thabassum.github.io	Figures 1 and 2 show the page loaded at that address	**Verified**
Stylesheet applied	Page displays with the dark theme and layout	Screenshots are fully styled; Sources panel lists style.css	**Verified**
Navigation anchors	Each of the 5 menu links points to an existing section	Script check: all 5 IDs exist in index.html	**Verified**
Desktop layout (1366 px)	Two-column hero, skills, projects and contact	Figure 1; test render: 2 columns in each grid	**Verified**
Tablet layout (768 px)	Two-column grids, no sideways scrolling	Test render only; no tablet screenshot supplied	**Verified**
Mobile layout (440 px and 375 px)	Single column, no sideways scrolling	Figure 2 and test render: 1 column, page width equals screen width	**Verified**
QR image loads	linkedin-qr.png appears in Contact section	Figure 2 Sources panel; test render shows image loaded	**Verified**
QR code content	QR leads to the author's LinkedIn profile	Decoded with OpenCV: a linkedin.com/in/ profile address	**Verified**
Project GitHub links	All project links open an existing repository	13 GitHub addresses in the page returned HTTP 200 on 16 Oct 2026; repository contents were not reviewed	**Verified**
External links open safely	New tab with rel="noopener noreferrer"	All 14 external links in the code have both attributes	**Verified**
"Connect on LinkedIn" button	Opens the author's own profile	Code links to https://www.linkedin.com/ (home page); a code comment says to replace it	**Needs fix**
Hover effects and smooth scroll	Cards lift, links change colour, scrolling is animated	Defined in CSS but not observed in the supplied screenshots	**Not verified**
Other browsers (Firefox, Safari, Edge)	Same appearance as Chrome	Only Chrome was used	**Not verified**
Real phone / tablet	Displays correctly on physical devices	Only browser emulation was used	**Not verified**
HTML/CSS validation and accessibility audit	No validator errors; adequate colour contrast	No validator or Lighthouse report supplied	**Not verified**

Summary: 10 checks verified, 1 item needs a fix (the LinkedIn button address) and 4 items are not verified. The page structure, responsive behaviour and links work as designed in the checks performed.

10. Advantages and Learning Outcomes

Advantages of the portfolio

- One link presents skills, education summary, twelve projects and contact options.
- Static HTML and CSS means fast loading, no server cost and free hosting on GitHub Pages.
- The responsive layout lets recruiters view it on a laptop or phone.
- Project cards link directly to source code on GitHub so that work can be inspected.
- The LinkedIn QR code makes it easy to connect from a phone.

HTML and CSS skills demonstrated

Area	Demonstrated in this project
HTML structure	Semantic page layout, internal anchor links, external links, images with alt text, meta tags
CSS layout	Flexbox and Grid for navigation, cards and page sections; sticky header
Visual design	Colour palette, gradients, shadows, rounded cards, consistent spacing and typography
Interaction	Hover transitions, keyframe animations and smooth scrolling without JavaScript
Responsive design	Viewport meta tag, `clamp()`, and breakpoints at 850 px and 650 px
Deployment	Publishing a static site with GitHub Pages and keeping files in a GitHub repository

11. Limitations and Future Enhancements

The following points describe current limitations and realistic next steps. None of the enhancements below is part of the current website.

Current limitation	Possible enhancement
The "Connect on LinkedIn" button points to the LinkedIn home page.	Replace the address with the author's profile URL (the same one the QR code uses).
No email address or phone number is displayed; there is no contact form.	Add an email link, or a form handled by a third-party service.
Education details are limited to one sentence in the About section.	Add a dedicated Education section and a downloadable resume.
Project cards show text and tags only.	Add screenshots, live demo links and short results for each project.
On small screens the navigation links stack under the logo, and the floating "Python" badge overlaps the code window's header dots (visible in Figures 2 and 3).	Use a collapsible menu and adjust badge positions at small widths.
Accessibility has not been audited.	Run Lighthouse or WAVE, check colour contrast of the muted text and test keyboard navigation.
Testing covered Chrome and emulation only.	Test in Firefox, Safari and Edge and on real phones; validate HTML and CSS.
Content is entirely hand-written HTML.	Add Open Graph tags for link previews and consider a light/dark theme option.

12. Conclusion

The Personal Portfolio Website meets the main goals of the internship project. Using only HTML5 and CSS3, it presents the author's introduction, background, skills and twelve projects in a clean, dark-themed, single-page design, and offers a LinkedIn QR code and GitHub link for contact. The layout adapts from desktop to mobile through CSS Grid, Flexbox and media queries, and the site is published online through GitHub Pages.

Building the project gave practical experience in structuring a page with semantic HTML, styling it with modern CSS, making it responsive and deploying it. The testing results show that the core behaviour was confirmed, while the open items (correct LinkedIn button address, cross-browser and accessibility testing) provide a clear plan for improvement.

13. References

[1] WHATWG. *HTML Living Standard*. https://html.spec.whatwg.org/

[2] W3C. *CSS - Cascading Style Sheets*. https://www.w3.org/Style/CSS/

[3] MDN Web Docs. *HTML: HyperText Markup Language*. https://developer.mozilla.org/en-US/docs/Web/HTML

[4] MDN Web Docs. *CSS: Cascading Style Sheets*. https://developer.mozilla.org/en-US/docs/Web/CSS

[5] GitHub Docs. *GitHub Pages documentation*. https://docs.github.com/en/pages

[6] Microsoft. *Visual Studio Code documentation*. https://code.visualstudio.com/docs

[7] Thabassum, S. *Personal Portfolio Website* (live site). https://suhana-thabassum.github.io/